

OPERATING SYSTEMS CONCEPT – LAB

EXP – 04: SYSTEM CALLS – fork(), exec(), getpid(), opendir(), readdir()

fork()

```
Activities Terminal Apr 20 18:16
rathnavelu06@localhost:~/rathna/systemCalls

[rathnavelu06@localhost systemCalls]$ vi fork.c
[rathnavelu06@localhost systemCalls]$ cat fork.c
#include <stdio.h>
#include <unistd.h>
int main(){
    if(fork()==0){
        printf("This is CHILD process\n");
    } else{
        printf("This is PARENT process\n");
    }
    printf("PID: %d\n", getpid());
    return 0;
}

[rathnavelu06@localhost systemCalls]$ gcc fork.c -o fork
[rathnavelu06@localhost systemCalls]$ ./fork
This is PARENT process
PID: 4267
This is CHILD process
PID: 4268
[rathnavelu06@localhost systemCalls]$
```

exec()

```
Activities Terminal Apr 20 18:18
rathnavelu06@localhost:~/rathna/systemCalls

[rathnavelu06@localhost systemCalls]$ vi exec.c
[rathnavelu06@localhost systemCalls]$ cat exec.c
#include <stdio.h>
#include <unistd.h>
int main(){
    printf("Before exec()\n");
    execl("/bin/ls", "ls", NULL);
    printf("After exec()\n");
    return 0;
}

[rathnavelu06@localhost systemCalls]$ gcc exec.c -o exec
[rathnavelu06@localhost systemCalls]$ ./exec
Before exec()
exe exec exec.c fork fork.c opendir opendir.c pid pid.c readdir readdir.c
[rathnavelu06@localhost systemCalls]$
```

getpid()

```
Activities Terminal Apr 20 18:19
rathnavelu06@localhost:~/rathna/systemCalls

[rathnavelu06@localhost systemCalls]$ vi pid.c
[rathnavelu06@localhost systemCalls]$ cat pid.c
#include <stdio.h>
#include <unistd.h>
int main(){
    printf("PID: %d\n", getpid());
    printf("PPID: %d\n", getppid());
    return 0;
}

[rathnavelu06@localhost systemCalls]$ gcc pid.c -o pid
[rathnavelu06@localhost systemCalls]$ ./pid
PID: 4421
PPID: 4376
[rathnavelu06@localhost systemCalls]$
```

opendir()

```
Activities Terminal Apr 20 18:21
rathnavelu06@localhost:~/rathna/systemCalls

[rathnavelu06@localhost systemCalls]$ vi opendir.c
[rathnavelu06@localhost systemCalls]$ cat opendir.c
#include <stdio.h>
#include <dirent.h>
int main(){
    DIR *dir;
    dir=opendir(".");
    if(dir==NULL){
        printf("Failed to open directory\n");
        return 1;
    }
    printf("Directory opened successfully!\n");
    closedir(dir);
    return 0;
}

[rathnavelu06@localhost systemCalls]$ gcc opendir.c -o opendir
[rathnavelu06@localhost systemCalls]$ ./opendir
Directory opened successfully!
[rathnavelu06@localhost systemCalls]$
```

readdir()

```
Activities Terminal Apr 20 18:22
rathnavelu06@localhost:~/rathna/systemCalls

[rathnavelu06@localhost systemCalls]$ vi readdir.c
[rathnavelu06@localhost systemCalls]$ cat readdir.c
#include <stdio.h>
#include <dirent.h>
int main(){
    DIR *dir;
    struct dirent *entry;
    dir=opendir(".");
    if(dir==NULL){
        printf("Cannot open directory\n");
        return 1;
    }
    printf("Files in directory:\n");
    while((entry=readdir(dir))!=NULL){
        printf("%s\n", entry->d_name);
    }
    closedir(dir);
    return 0;
}

[rathnavelu06@localhost systemCalls]$ gcc readdir.c -o readdir
[rathnavelu06@localhost systemCalls]$ ./readdir
Files in directory:
.
..
pid.c
opendir.c
readdir.c
fork.c
fork
exec.c
exe
exec
pid
opendir
readdir
[rathnavelu06@localhost systemCalls]$
```